ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ROBOTICS FUNDAMENTALS

PAPER 1 · PRACTICE D · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Robotics Fundamentals paper 1 · practice d.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers and wording.

SECTION A

QUESTION 1

1 1.1 C ✓ 1.2 A ✓ 1.3 B ✓ 1.4 D ✓ 1.5 A ✓ 1.6 C ✓ 1.7 B ✓ 1.8 D ✓ 1.9 A ✓ 1.10 B ✓ (10)

[10]

QUESTION 2

2 2.1 False ✓ 2.2 True ✓ 2.3 True ✓ 2.4 False ✓ 2.5 True ✓ 2.6 False ✓ 2.7 True ✓ 2.8 True ✓ 2.9 False ✓ 2.10 True ✓ (10)

[10]

QUESTION 3

3 3.1 C ✓ 3.2 G ✓ 3.3 J ✓ 3.4 B ✓ 3.5 E ✓ 3.6 D ✓ 3.7 I ✓ 3.8 F ✓ 3.9 A ✓ 3.10 H ✓ (10)

[10]

QUESTION 4

4 4.1 variable ✓ 4.2 light ✓ 4.3 cathode ✓ 4.4 steps ✓ 4.5 digital ✓ (5)

[5]

TOTAL SECTION A: [35]

SECTION B

QUESTION 5

5.1 Any FIVE ✓✓✓✓✓: frame; print bed (heated bed); extruder / hot end and nozzle; stepper motors; filament spool and holder; control board; display / control knob; power supply; belts and rods. (5)

5.2 The amount (percentage) and pattern of material printed inside the part ✓; more infill makes the part stronger and heavier ✓ but uses more filament and takes longer to print ✓. (3)

5.3 Any FOUR ✓✓✓✓: do not touch the hot nozzle or heated bed; keep hands away from moving parts while printing; print in a ventilated area (fumes, especially ABS); never leave a print running unattended for long periods; switch off and let it cool before cleaning or changing parts; keep flammable materials away; use the correct power supply. (4)

[12]

QUESTION 6

6.1 gpiozero ✓; it gives simple ready-made classes (LED, Button) so you do not have to set up pins and modes yourself ✓. (2)

6.2 green = LED(17) creates an LED object on GPIO 17 (set up as an output) ✓✓; start_button = Button(2) creates a button object on GPIO 2 (input with a pull-up) ✓. (3)

6.3 Whether the conveyor should be running (True) or stopped (False) ✓; line 14 sets it to True ✓ and line 16 sets it to False ✓. (3)

6.4 True: green LED on ✓ and red LED off ✓. False: green off and red on ✓. (3)

6.5 About 5 messages ✓✓, because each time round the loop while running there is a 1-second sleep ✓. (3)

6.6 So the loop does not run thousands of times a second and keep the processor busy ✓✓, while still checking the buttons often enough to react quickly ✓. (3)

6.7 running is first set to True (line 14) but then set to False (line 16) ✓✓, so the STOP button wins and the conveyor stops – a safe design ✓. (3)

6.8 Use the Buzzer class (from gpiozero import Buzzer) ✓; buzzer = Buzzer(22) ✓; buzzer.on() in the else part and buzzer.off() when running ✓. (3)

[23]

QUESTION 7

7.1 The rate of flow of electric charge (electrons) through a conductor ✓, measured in amperes ✓. (2)

7.2 $\mathrm{I} = \mathrm{V} \div \mathrm{R}$ ✓ $= 12 \div 4$ ✓ $= 3\ \mathrm{A}$ ✓ (3)

7.3 $\mathrm{P} = \mathrm{V} \times \mathrm{I}$ ✓ $= 12 \times 3$ ✓ $= 36\ \mathrm{W}$ ✓ (3)

7.4 $\mathrm{E} = \mathrm{P} \times \mathrm{t} = 36 \times 2$ ✓ $= 72\ \mathrm{Wh}$ ✓ $= 0{,}072\ \mathrm{kWh}$ ✓ (3)

7.5 $\mathrm{R_T} = (6 \times 3) \div (6 + 3)$ ✓✓ $= 2\ \Omega$ ✓ (3)

7.6 $\mathrm{I} = 6 \div 2$ ✓ $= 3\ \mathrm{A}$ ✓ (2)

7.7 An open circuit has a break in it, so no current flows ✓✓ (e.g. a broken wire or open switch). A short circuit is a very low-resistance path that lets a very large current flow, bypassing the load ✓✓, which can cause overheating or fire. (4)

[20]

QUESTION 8

8.1 A robot must protect its own existence ✓ as long as such protection does not conflict with the First ✓ or Second Law ✓. (3)

8.2 Yes – the robot should go ✓. The First Law says a robot may not, through inaction, allow a human to come to harm ✓, so helping the trapped miner is the most important duty ✓. The Second Law says it must obey human orders ✓, and the supervisor's order does not harm anyone ✓. The Third Law (protecting itself) only applies when it does not conflict with the First and Second Laws ✓, so the risk of damage to the robot does not stop it from going ✓. (7)

[10]

TOTAL SECTION B: [65]

TOTAL: 100

END OF MARKING GUIDELINE